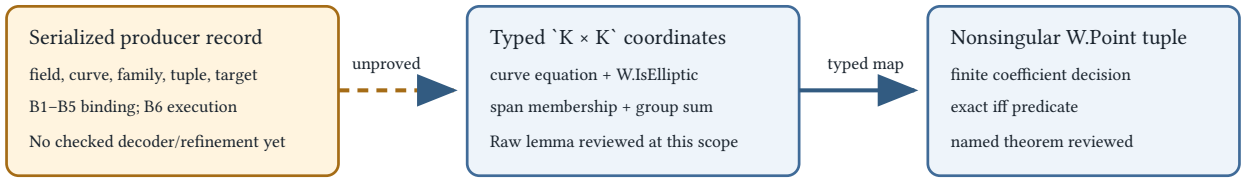
Semaev decomposition certificate: independent fidelity review

8 November 2026 · Claim THM-SEMAEV-FALL1-LEMMA-3 · review source 6a0eb966

The blind mathematical reviewer derived the finite span decision and exact iff predicate independently. The source reviewer matched the named Lean theorem to a typed nonsingular-point predicate and the supplemental raw lemma to typed field coordinates under `W.IsElliptic`. A historical hosted sweep has an exact-target receipt, and 122 archived input bindings passed fresh consistency checks. The forced current-pin Lean run 42446577502 was pending when this report was compiled; its result must be read back before a fresh-machine pass is assigned.

Semaev decomposition certificate: checked mathematical scope and input gap



J1: independent derivation and 72 F5 controls · J2: source/type fidelity and six gaps · J3: archived receipt consistency; fresh Lean run pending

Review source 6a0eb966 · theorem source SHA-256 aa022e48… · TASK-20261108-095e46 / f75797 / 5ef48a

Separate verdicts

Claim boundary	Machine	Fidelity	Outcome
Typed finite decomposition	Historical receipt consistent; fresh run pending	Faithful for nonsingular W.Point	Inconclusive pending fresh run and independence repair
Raw K×K coordinates	Source inspected; fresh raw controls pending	Faithful under W.IsElliptic	Inconclusive pending controls
Serialized producer record	No checked decoder or native refinement	Partial; B1–B6 missing	Needs revision

Evidence and limits

- J1 (TASK-20261108-095e46): blind card only; exact finite enumeration, empty basis/tuple cases, and 72 F5 controls (14 accepted, 58 rejected, zero mismatches).
- J2 (TASK-20261108-f75797): six pinned claim/spec/source files and pinned Mathlib sources matched. The serialized-record bridge requires parsing, field interpretation, curve/ellipticity, span, tuple/target, and authenticated execution or native-code refinement.
- J3 (TASK-20261108-5ef48a): 122 input rows matched; GitHub artifact digest authenticated; 13 receipt rejection controls passed. A consistent synthetic receipt was accepted, confirming that receipt validation alone is not execution authentication. PR 3173's green formal-workspace check skipped the proof workspace.

The independence checker failed on a prospective-plan and attestation schema mismatch: short versus full joint labels, a JSON blind attestation not discovered as YAML, and a scalar `blind_from` where protected producer paths must be listed. Its failure is retained verbatim. Sealed reviewer artifacts and the prospective plan were not rewritten. This round therefore does not claim `validated_scoped`.

The archived verifier input map also omits its imported `orchestration/fast_yaml.py` helper, separately pinned by J3. A complete transitive software-input claim requires that closure to be repaired.

Next construction

Bind serialized bytes to the concrete field, curve, family, tuple, and target; prove/check the decoder bridge and execute the instantiated predicate on a pinned record. Repair the verifier input closure and run a new prospective independence round with checker-compatible bindings. No hypothesis or official evidence status changes from this technical review.

Full source-linked technical review, failed-check transcript, pinned inputs, and sealed reviewer reports are in this directory.